What is OOPS?

OOPS means:

Object Oriented Programming System

It is a way of writing programs using **objects**.

Java uses OOPS because real-world things are easier to understand using objects.

First Understand “Object”

Everything around you is an object.

Examples:

- Car
- Mobile
- TV
- Dog
- Baby
- Student

Every object has:

Thing	Real Meaning
Properties	Data / characteristics
Actions	What it can do

Example:

Baby Object 👶

Properties (data)

- name
- age
- color
- weight

Actions (behavior)

- cry()
- sleep()
- drinkMilk()

Same Thing in Java

```
class Baby {

  String name;
  int age;

  void cry() {
    System.out.println("Baby is crying");
  }

  void sleep() {
    System.out.println("Baby is sleeping");
  }
}
```

Creating object:

```
public class Main {
  public static void main(String[] args) {
```

```
        Baby b1 = new Baby();

        b1.name = "Ali";
        b1.age = 1;

        b1.cry();
    }
}
```

Why Java Uses OOPS?

Because OOPS makes software:

- easy to build
- easy to understand
- easy to reuse
- easy to maintain

Imagine making:

- Instagram
- WhatsApp
- Amazon
- Banking App

Without OOPS it becomes a giant messy file 😵

With OOPS:

- User → separate object
- Message → separate object
- Payment → separate object
- Product → separate object

Everything becomes organized.

Real Life Example — Car Factory 🚗

Imagine a car company.

Without OOPS:

One huge file with everything mixed:
engine code
door code
music code
brake code

Very confusing.

With OOPS:

```
Car object
 ├── Engine object
 ├── Brake object
 ├── Door object
 └── MusicSystem object
```

Everything is separated properly.

That is why Java uses OOPS.

Advantages of OOPS

1. Reusability ♻️

Write once → use many times.

Example:

```
class Animal {
   void eat() {
      System.out.println("Eating");
   }
}
```

Dog can reuse it.

```
class Dog extends Animal {
}
```

Now Dog already has eat().

No need to write again.

This saves time.

Real Example

Your father buys one TV remote.

Everyone in house reuses it.

No need to create separate remote for each person.

That is reuse.

2. Easy Maintenance

If bug comes:

You fix only one class.

Not whole project.

Real Example

If fan switch is broken:

Electrician fixes only switch.

Not entire house wiring.

3. Security

OOPS hides important data.

This is called **Encapsulation**.

Example ATM:

You can withdraw money.

But cannot directly touch bank database.

Real Example

Mobile phone:

You press buttons.

But internal motherboard is hidden.

4. Easy to Understand 📚

Code looks like real world.

Example:

Car car1 = new Car();
Student s1 = new Student();

Very natural.

5. Flexibility 🔄

Same thing can behave differently.

Example:

Animal a;

a = new Dog();
a = new Cat();

This is polymorphism.

Real Example

One person can be:

- Father at home
- Employee at office
- Customer in shop

Same person → different behavior.

6. Faster Development 🚀

Teams can work separately.

Example:

Developer	Work
Dev 1	Login
Dev 2	Payment
Dev 3	Profile
Dev 4	Notifications

Everything combines later.

Four Main Pillars of OOPS

Pillar	Simple Meaning
Encapsulation	Hide data
Inheritance	Reuse code
Polymorphism	Many forms
Abstraction	Show only important things
Interface	Only the design

Super Simple Real Example — Mobile 📱

Encapsulation

You use apps.

Internal hardware hidden.

Inheritance

Samsung phone inherits Android features.

Polymorphism

Power button:

- short press → lock
- long press → restart

Same button → different behavior.

Abstraction

You drive car using steering.

No need to know engine internals.

Final Easy Definition

OOPS

OOPS is a programming style where we create software using objects similar to real-world things.

Java uses OOPS because it makes programs:

- organized
- reusable
- secure
- scalable

- easy to maintain